ProjectDora

KOSGEB Data Orchestration Platform

Architecture Reference Document

Version 1.0 — March 2026 | Beta Release

1. System Overview

ProjectDora is a **modular monolith** built on Orchard Core CMS 2.1.4 and .NET 10. It serves KOSGEB (Turkey's SME development agency) with content management, workflow automation, query management, multi-tenant isolation, and integration capabilities. All 10 custom modules compile into a single .NET process with boundaries enforced via interface contracts in ProjectDora.Core.Abstractions.

ADR	Decision	Rationale
ADR-001	Modular Monolith	Single process, module contracts via IContentService etc.
ADR-002	Orchard Core 2.1.4	YesSql, Recipes, Tenants, OpenId, Workflows built-in
ADR-003	Clean Architecture	Core.Abstractions layer isolated from OC implementation
ADR-005	PostgreSQL Primary	3 schemas: orchard, audit, analytics — no cross-schema SQL
ADR-006	Redis Cache (optional)	OrchardCacheService degrades to placeholder when absent
ADR-007	Lucene (dev) / ES (prod)	LuceneIndexRebuilderAdapter wraps OC concrete classes
ADR-008	OpenID Connect	OC OpenId module — JWT bearer for API, cookie for admin UI
ADR-010	Multi-Tenant Isolation	Shell-level — each tenant: separate schema or SQLite file

2. Module Architecture (10 Custom Modules)

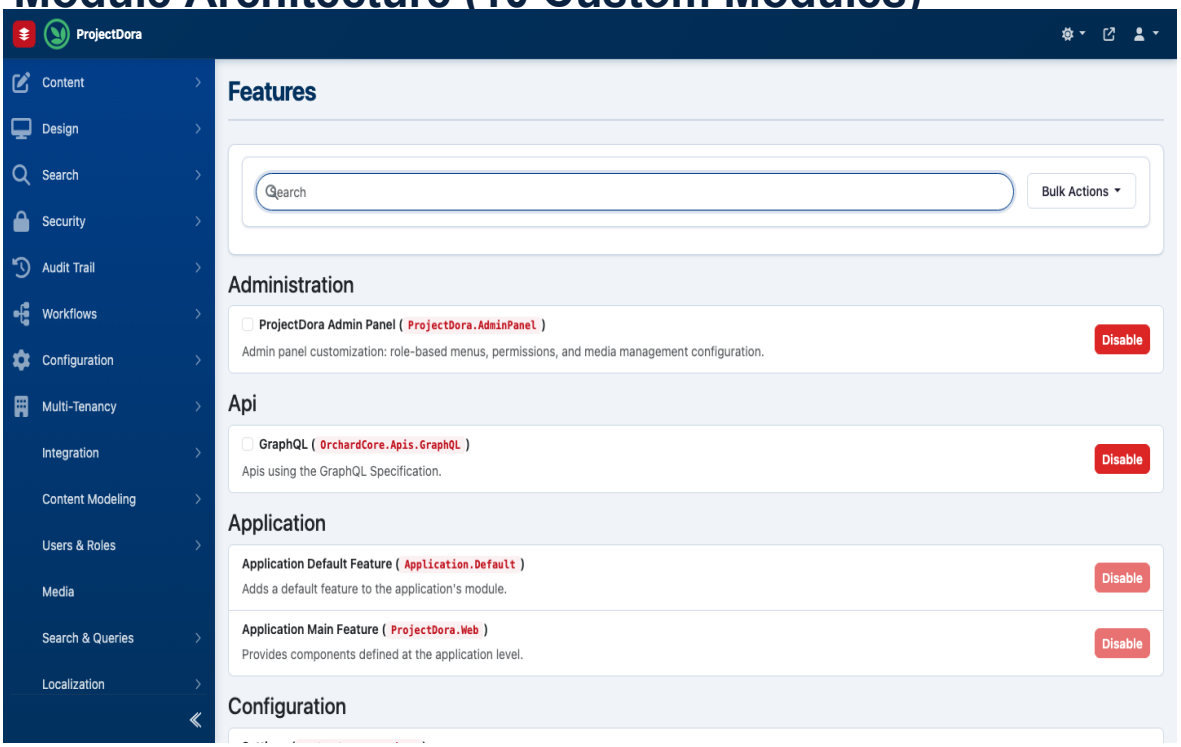

Figure 2.1 — All 10 ProjectDora modules enabled in Feature Management

Module	Spec §	Key Responsibility
ProjectDora.AdminPanel	4.1.1	KOSGEB branding, role-based menus, media config, RBAC UI
ProjectDora.ContentModeling	4.1.2	Dynamic types, parts, fields — wraps OC ContentDefinition
ProjectDora.QueryEngine	4.1.5	Lucene/ES/SQL queries, LuceneIndexRebuilderAdapter, parameterized API
ProjectDora.UserManagement	4.1.6	Users, roles, GeneratedPermissionProvider → OC auth pipeline
ProjectDora.Workflows	4.1.7	OrchardWorkflowService, real executionId, trace: markers
ProjectDora.Localization	4.1.8	CulturePicker, LocalizationPart, PO file management
ProjectDora.AuditTrail	4.1.9	Versioned diffs, retention, audit event indexing
ProjectDora.Infrastructure	4.1.10	Multi-tenant, Redis metrics, recipes, OpenID config
ProjectDora.Integration	4.1.11	Webhooks+delivery log+retry, ApiEndpoints, GraphQL
ProjectDora.ThemeManagement	4.1.4	Liquid templates, Monaco editor, KOSGEB admin CSS theme

3. Runtime Architecture

The layered architecture enforces dependency inversion at the Core.Abstractions boundary. All Orchard Core dependencies are wrapped by concrete service adapters.

Layer	Components
HTTP Clients	Admin browser, external REST/GraphQL API consumers
ASP.NET Core Pipeline	Middleware: Auth, CORS, Rate Limiting, OC Shell multi-tenant routing
Admin Controllers	WebhooksController, ApiEndpointsController — [Authorize] + permission check
Service Interfaces	IContentService, IWorkflowService, ICacheService, IIntegrationService
OC Adapters	OrchardWorkflowService, OrchardCacheService, OrchardIntegrationService
Data Layer	YesSql (PostgreSQL/SQLite), EF Core (audit), Redis, Lucene/ES

Component	Technology	Notes
Runtime	.NET 10 (C#)	Single process, all modules co-deployed
CMS Framework	Orchard Core 2.1.4	YesSql documents, shell tenants, OpenId, Workflows
Primary DB	PostgreSQL 16	YesSql + EF Core audit schema + analytics schema
Dev DB	SQLite	Embedded, no external dependency for local development
Cache	Redis 7 (StackExchange.Redis)	Optional — graceful degradation to zero stats
Search	Lucene.NET (dev) / Elasticsearch 8 (prod)	LuceneIndexRebuilderAdapter wraps concrete OC classes
Object Storage	MinIO (S3-compatible)	Media files — port 9000/9001
Auth	OpenID Connect (OC OpenId module)	JWT bearer for API clients, cookie for admin UI
Logging	Serilog	Structured JSON, sink to stdout + file
Observability	OpenTelemetry + Grafana/Prometheus	Metrics, distributed traces, dashboards
Containers	Docker / Docker Compose	Services: web, postgres, redis, elasticsearch, minio

4. Security Architecture

Security is enforced at multiple layers. The beta hardening work connected the **GeneratedPermissionProvider** to OC's authorization pipeline, ensuring content-type-level RBAC functions end-to-end.

Permission Resolution Flow:

1. Static permissions declared in Permissions.cs per module → IPermissionProvider.GetPermissionsAsync()
2. Dynamic content-type permissions: OrchardRoleService.GenerateContentTypePermissionsAsync() → stores to GeneratedPermissionsData (ISiteService)
3. **GeneratedPermissionProvider** reads GeneratedPermissionsData → feeds into IPermissionProvider chain
4. IAuthorizationService.AuthorizeAsync(user, permission) evaluates both static + dynamic permissions
5. Controller guards: [Authorize] + if (!await _authService.AuthorizeAsync(User, Permissions.X)) return Forbid()

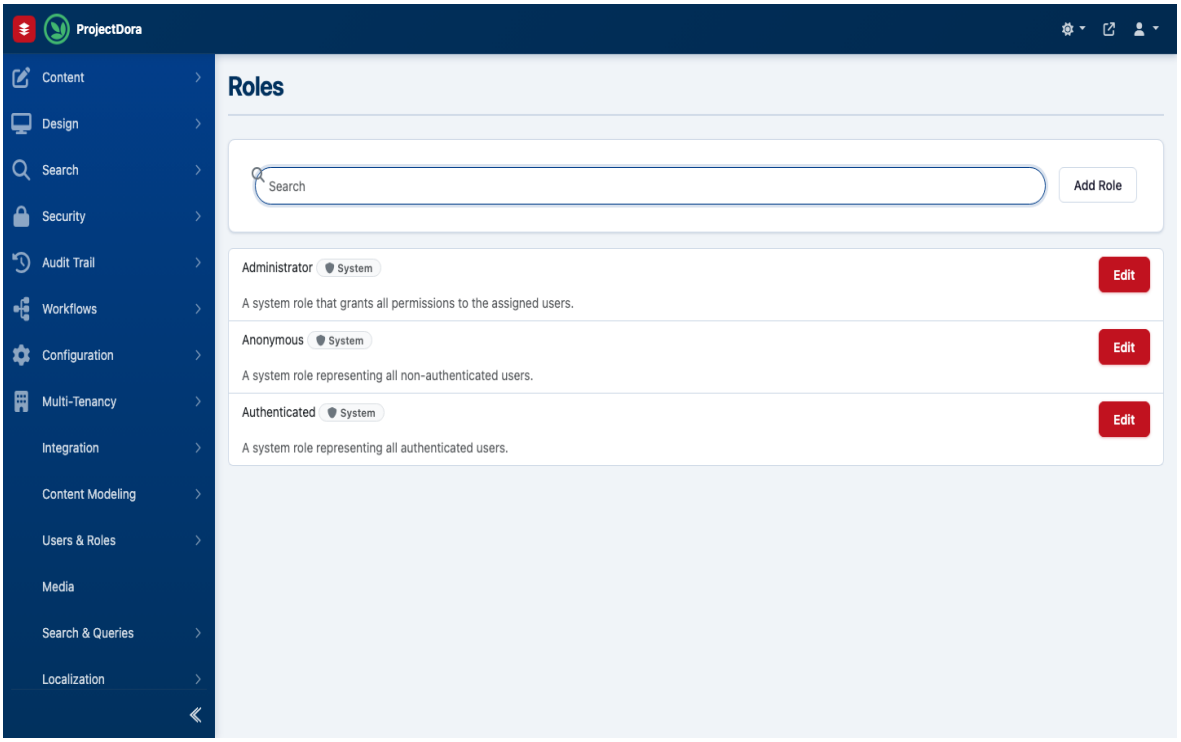

Figure 4.1 — Roles Management (RBAC)

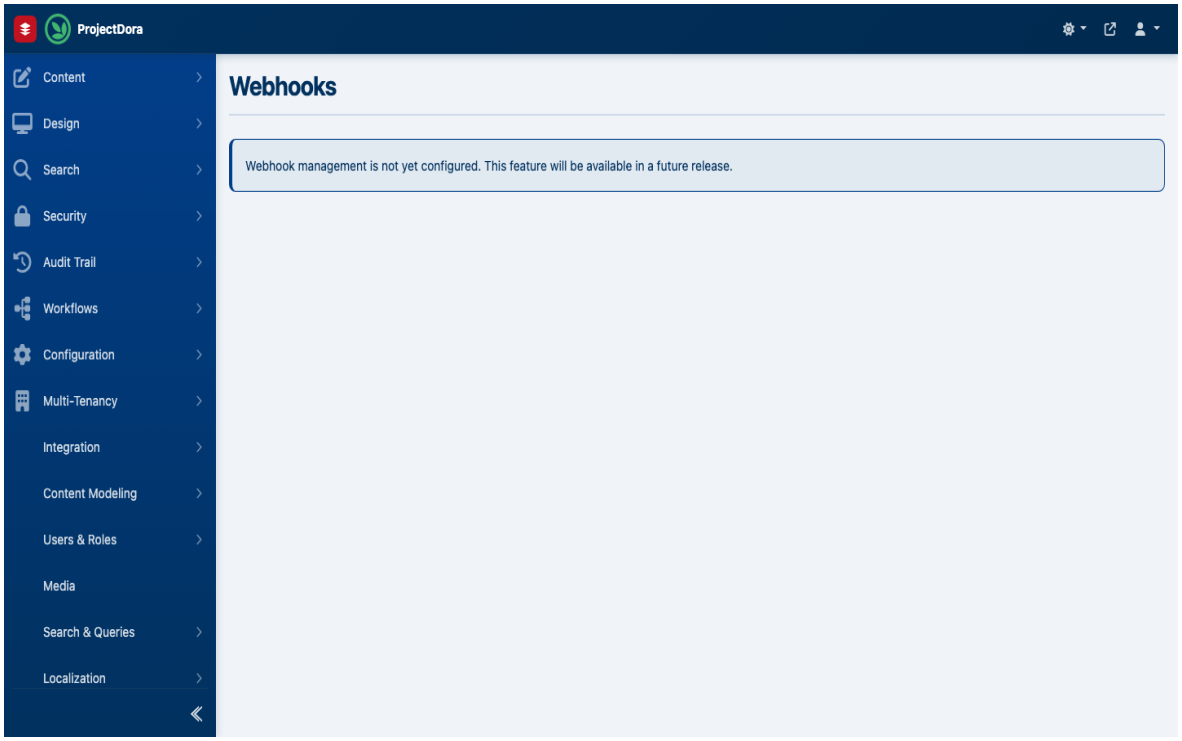

Figure 4.2 — Webhooks (permission-guarded controller)

5. Key Service Implementations (Beta Fixes)

5.1 OrchardWorkflowService — Execution ID Tracking

Scenario	Return Value	Caller Behavior
OC creates persisted Workflow record	Workflow.Id.ToString() — numeric, queryable	GetExecutionAsync(id) returns the record
Signal matched no active handlers	"trace:{workflowId}:{unixMs}" prefix	Caller must check trace: prefix; skip GetExecutionAsync
WorkflowType not found	throws KeyNotFoundException(*999*)	Caller handles exception
WorkflowType.IsEnabled == false	throws InvalidOperationException(*disabled*)	Caller handles exception

5.2 GeneratedPermissionProvider

Implements IPermissionProvider. Reads GeneratedPermissionsData.PermissionNames from ISiteService.GetSiteSettingsAsync(). Maps each name to Permission(name, name) { Category = 'Generated' }. Registered: services.AddScoped<IPermissionProvider, GeneratedPermissionProvider>() in UserManagement Startup.cs.

5.3 LuceneIndexRebuilderAdapter

Wraps OC's concrete LuceneIndexSettingsService and LuceneIndexingService (no interfaces exist). ListIndexNames(): GetSettingsAsync().Select(s => s.IndexName). RebuildAsync(name): _indexingService.RebuildIndexAsync(name). Registered in QueryEngine Startup.cs; OrchardCore.Search.Lucene 2.1.4 added to .csproj.

5.4 OrchardCacheService — Redis Metrics

Constructor resolves IConnectionMultiplexer? via services.GetService(typeof(IConnectionMultiplexer)). Redis present: server.InfoAsync() returns IGrouping<string,KVP<string,string>>[] — parse keyspace_hits, keyspace_misses, used_memory from named sections. Redis absent: returns CacheStatsDto(totalKeys, 0L, 0L, 0.0, 0L).

5.5 WebhookDeliveryLog (OrchardIntegrationService)

Delivery attempts persisted via ISiteService (IntegrationSettings.DeliveryLogs dictionary). Max 10 entries per webhook (FIFO — oldest discarded on overflow). Each entry: webhookId, url, statusCode, success, attemptedAt, wasRetry. On HTTP 5xx or timeout: 1x immediate retry → both attempts logged (wasRetry=false/true).

6. Data Architecture

Schema	ORM / Store	Key Tables / Collections
orchard	YesSql (document store)	Document (Content, Users, Settings, Workflows, Tenants), UserIndex, WorkflowIndex, Co
audit	EF Core (custom)	AuditTrailEventIndex — NormalizedUserName (UPPERCASE), EventName, CreatedUtc,
analytics	EF Core (planned)	Read-only aggregates — ETL pipeline from orchard schema

Critical OC API Notes:

API / Type	Correct Usage
IWorkflowTypeStore	WorkflowType definitions — GetAsync(long id), ListAsync()
IWorkflowStore	Execution instances — ListAsync(string workflowTypeId, int? skip, int? take)
AuditTrailEventIndex	Has NormalizedUserName (NOT UserName) — use .ToUpperInvariant() to match
ISite.Alter<T>()	Requires using OrchardCore.Entities (extension on IEntity)
IAuthorizationService	OC extension: AuthorizeAsync(user, Permission) — returns bool (no .Succeeded)
IRecipeExecutor	ExecuteAsync() returns Task<string> (execution ID, not Task)
ISession.Save()	Obsolete — use await session.SaveAsync()

7. Test Architecture — 344 Tests Passing

Project	Count	Scope
ProjectDora.Core.Tests	3	Core abstractions, DTOs, command contracts
ProjectDora.Modules.Tests	341	Unit (Moq), Integration (controller guards), UAT (E2E smoke)
Total	344	0 failures, 0 skipped — dotnet test ProjectDora.sln

Category	[Trait]	Example Classes
Unit	Category=Unit	WorkflowExecutionServiceTests, CacheServiceTests, RolePermissionTests
Integration	Category=Integration	PermissionGuardTests — controller Forbid/View assertions
UAT	Category=Uat	WorkflowTriggerTests, TenantLifecycleTests, ContentLifecycleTests, PermissionEnforcement
Security	Category=Security	SqlInjectionTests, PermissionSecurityTests

8. Deployment Architecture

Container	Image	Port	Notes
projectdora-web	.NET 10 / OC 2.1.4	5000:8080	Healthcheck: /health/live — waits for postgres+redis
projectdora-postgres	postgres:16	5432	Init: init-schemas.sql — creates orchard/audit/analytics
projectdora-redis	redis:7	6379	Optional — app degrades gracefully without it
elasticsearch	elasticsearch:8	9200	Production search; Lucene used in dev/small deployments
minio	minio/minio	9000/9001	Media object storage (S3-compatible)

Key Environment Variables:

```
ConnectionStrings__Default=Host=postgres;Port=5432;Database=projectdora;Username=...;Password=...
Redis__ConnectionString=redis:6379 Elasticsearch__Url=http://elasticsearch:9200 MinIO__Endpoint=minio:9000 |
MinIO__AccessKey=... | MinIO__SecretKey=... ASPNETCORE_ENVIRONMENT=Development | Production
```

9. Admin Interface Reference

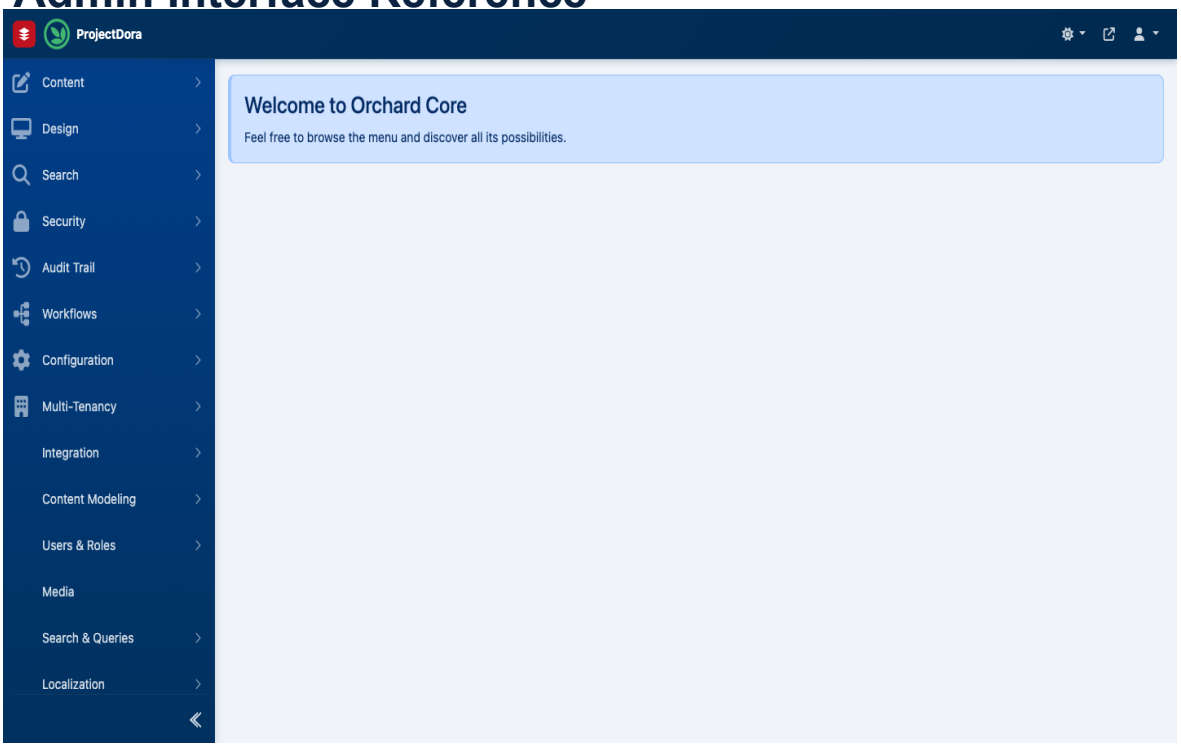

Figure 9.1 — Admin Dashboard (KOSGEB Navy Theme)

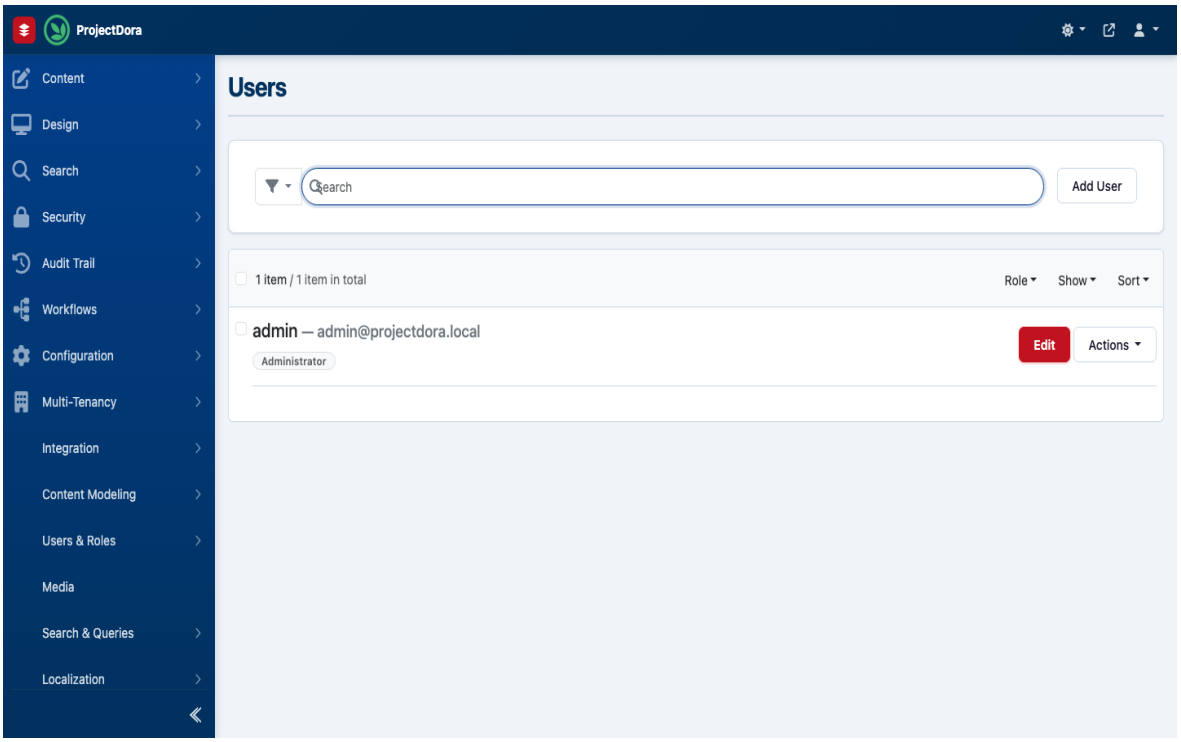

Figure 9.2 — User Management

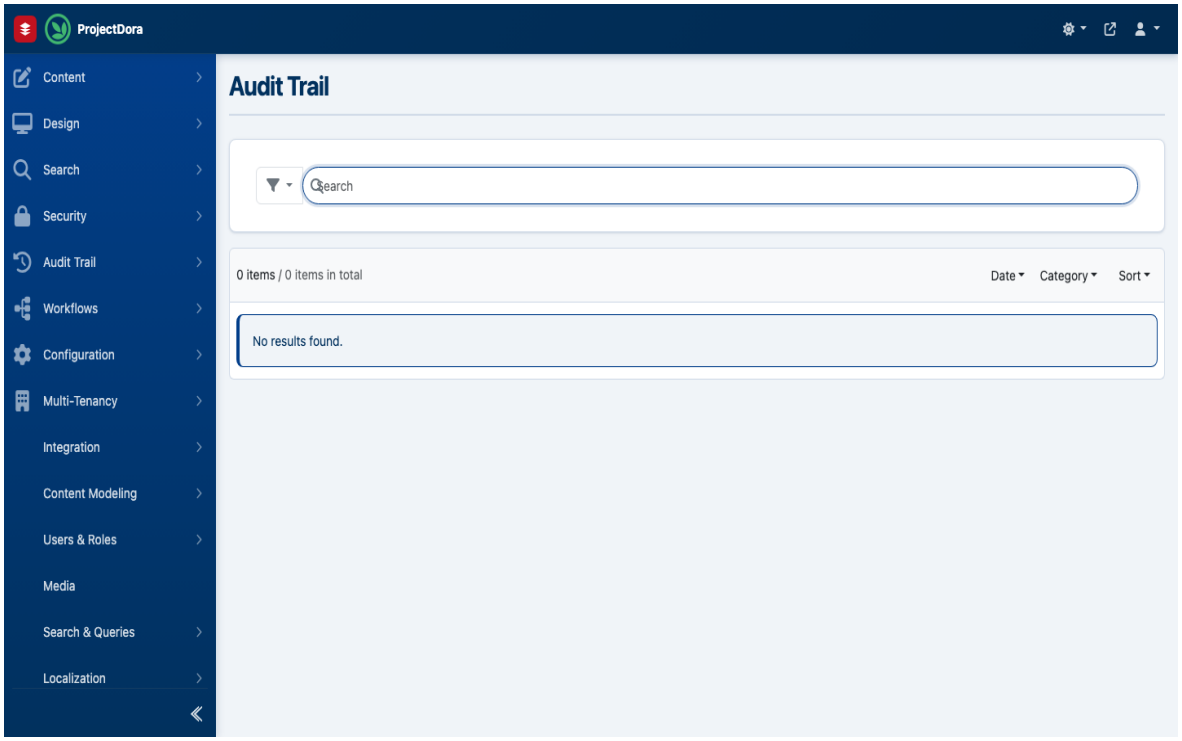

Figure 9.3 — Audit Trail

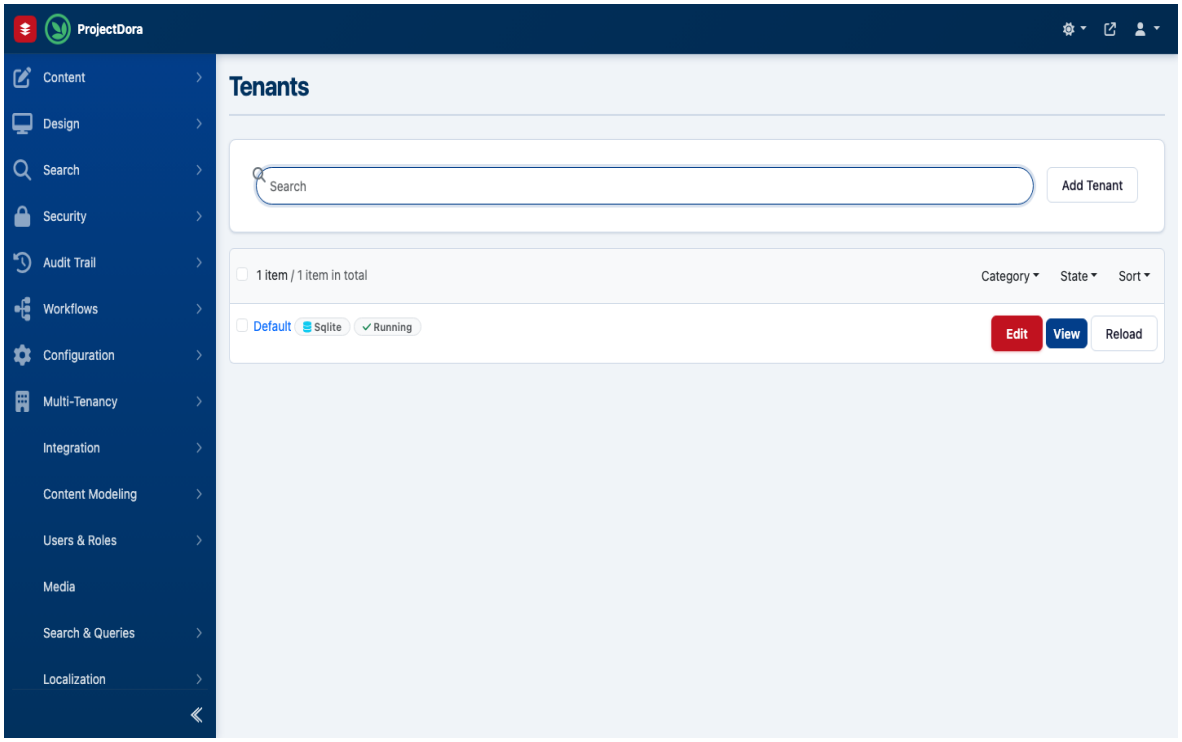

Figure 9.4 — Multi-Tenancy Management

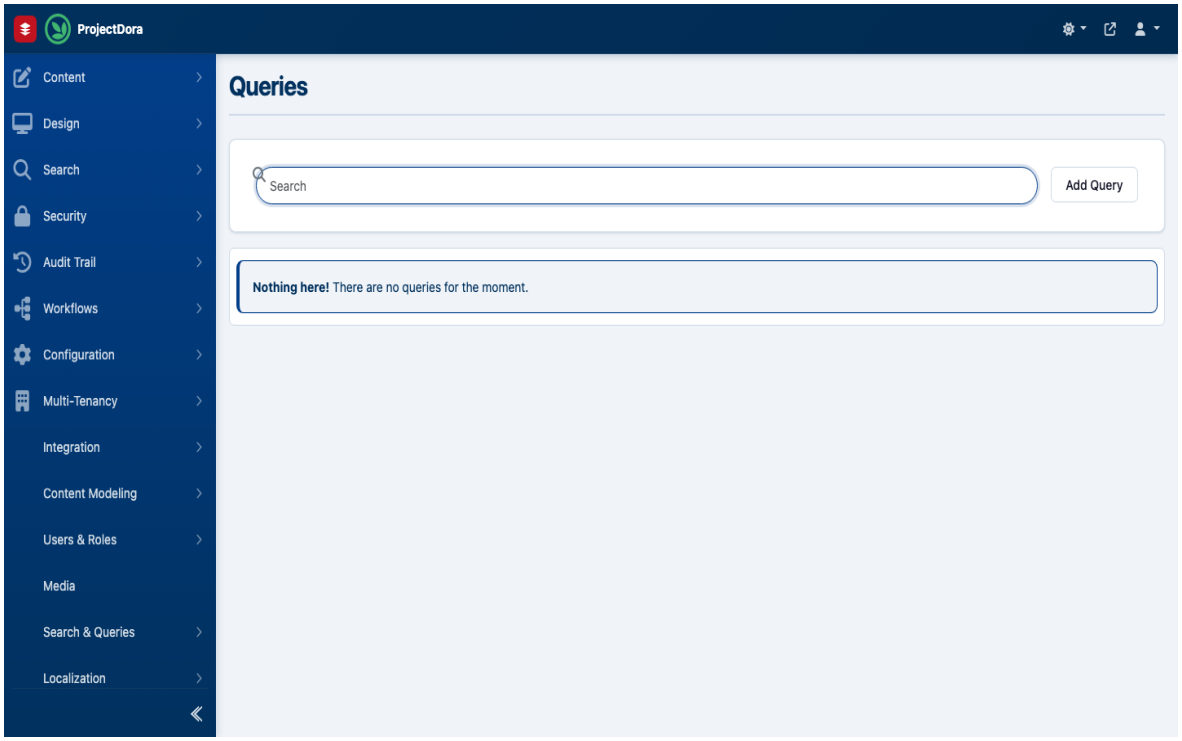

Figure 9.5 — Query Engine

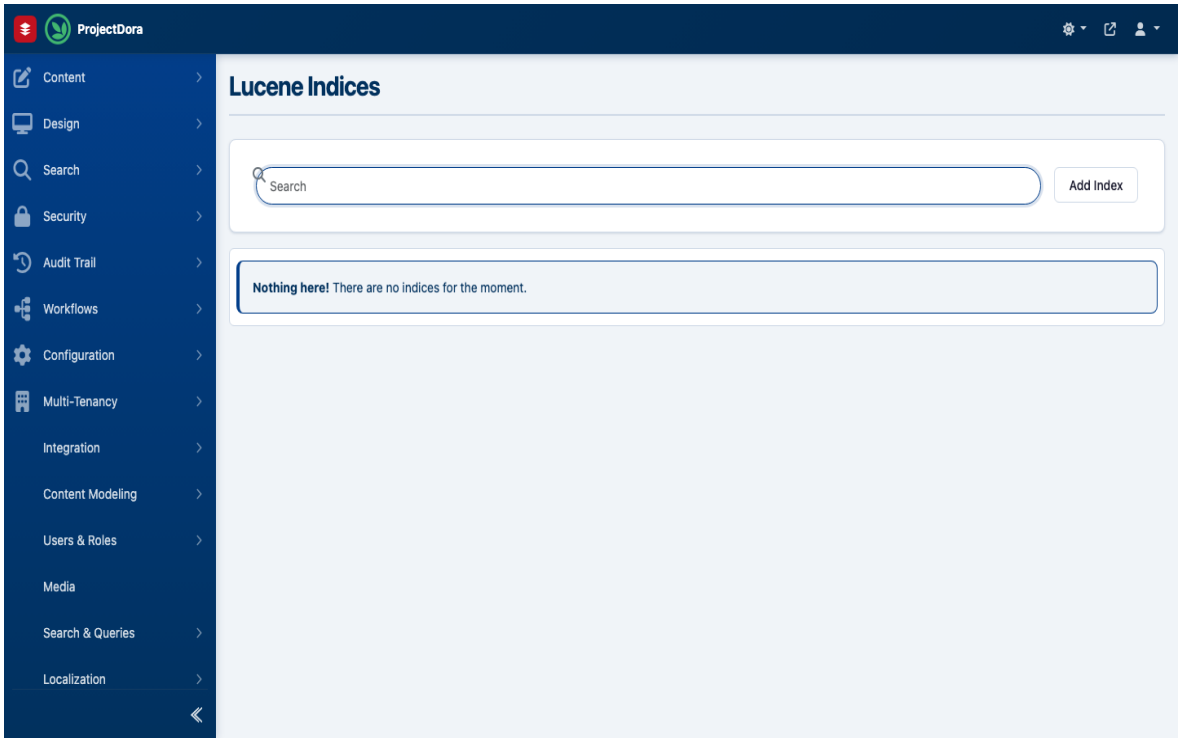

Figure 9.6 — Lucene Search Indices

10. Beta Release Status

ID	Issue	Status	Fix Summary
P0-1	GeneratedPermissionProvider missing	FIXED	New class implements IPermissionProvider; reads GeneratedPermissionsData;
P0-2	LuceneIndexRebuilder not in DI	FIXED	LuceneIndexRebuilderAdapter wraps concrete OC classes; registered in Query
P0-3	Manifest.cs missing 5 modules	FIXED	Added to: Localization, AuditTrail, Infrastructure, Integration, ThemeManageme
P1-1	Workflow executionId was fake UUID	FIXED	Returns real Workflow.Id; trace: marker for non-persisted signals
P1-2	Webhook no delivery log	FIXED	DeliveryLogs persisted; 1x retry on 5xx; max 10 entries per webhook
P1-3	API controllers unguarded	FIXED	WebhooksController + ApiEndpointsController use IAuthorizationService checks
P1-4	Cache stats always zero	FIXED	Optional IConnectionMultiplexer; real INFO stats when Redis available
P2-1	E2E smoke tests missing	DONE	+16 UAT tests: WorkflowTrigger, TenantLifecycle, ContentLifecycle, Permission
P2-2	No deployment seed data	DONE	demo-setup.json recipe: seed roles, users, DestekProgrami content type, Lucer
P2-3	No beta acceptance criteria	DONE	docs/beta-acceptance-checklist.md: P0-P4 criteria, release gate decision table

Beta Release Verdict: GO